Real Analysis II - MAT 2305

No of Credits - 03

Prerequisiste - MAT 1205

Contents ;

Chapter 01 - Infinite Series

Chapter 02 - Power Series

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Chapter 04 - Functions of Several variables [Long]

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Grading System ;

Tutorials = 20%

Mid Exam = 20%

End Exam = 60% (3 hours) Q_1, Q_2, Q_3, Q_4, Q_5, Q_6.

Chapter 01 ; INFINITE SERIES

$\mathbb{N} = \{1, 2, 3, \ldots\}$ - The set of all natural number.

$\{a_n\} \leftarrow$ Sequence

$(S_n) \leftarrow$ A new Sequence from $\{a_n\}$

$S_1 = a_1$

$S_2 = a_1 + a_2$

$S_3 = a_1 + a_2 + a_3$

$\vdots$

$S_n = a_1 + a_2 + a_3 + \ldots + a_n$

Let $a_n = \frac{1}{n}$, $n \in \mathbb{N}$

$S_1 = a_1 = \frac{1}{n} = 1$

$S_2 = a_1 + a_2 = 1/1 + 1/2 = 3/2$

$S_3 = a_1 + a_2 + a_3 = 1/1 + 1/2 + 1/3 = 11/6$

Let (a_n) be a sequence of real number. we associate in other Sequence (S_n) with (a_n) Such that,

$$S_n = a_1 + a_2 + a_3 + \ldots + a_n$$
$$= \sum_{k=1}^{n} a_k \quad \forall\, n \in \mathbb{N}$$

The sequence S_n is Called an infinite Series (or just a Series) and it is denoted by the expression $\sum_{n=1}^{\infty} a_n$.

$(S_n) = a_1, a_1 + a_2, a_1 + a_2 + a_3, \ldots\ldots, \sum_{k=1}^{n} a_k, \ldots \leftarrow$ Series

S_n is called the n^{th} partial sum of the infinite series.

The infinite series $\sum_{n=1}^{\infty} a_n$ is said to be convergent or be divergent.

If the sequence (S_n) converges to S, then S is called the sum of the infinite series. and we write.

$$\sum_{n=1}^{\infty} a_n = S \longleftarrow \text{A unique value.}$$

$$(S_n) = \left(a_1, \sum_{k=1}^{2} a_k, \sum_{k=1}^{3} a_k, \sum_{k=1}^{4} a_k, \ldots, \sum_{k=1}^{n} a_k, \sum_{k=1}^{n+1} a_k, \ldots\right)$$

i.e.

$$\boxed{\sum_{n=1}^{\infty} a_n = S = \lim_{n\to\infty} S_n = \lim_{n\to\infty} \sum_{k=1}^{n} a_k}$$

Ex 01 :- Prove that $\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1$

Let $a_n = \frac{1}{n(n+1)}$ $\forall n \in \mathbb{N}$

$S_1 = a_1 = \frac{1}{1.2}$

$S_2 = a_1 + a_2 = \frac{1}{1.2} + \frac{1}{1.3}$

$S_3 = a_1 + a_2 + a_3 = \frac{1}{1.2} + \frac{1}{1.3} + \frac{1}{1.4}$

Then $S_n = a_1 + a_2 + a_3 + \ldots + a_n = \sum_{k=1}^{n} a_k$

$$= \sum_{k=1}^{n} \frac{1}{k(k+1)}$$

$a_n = \frac{1}{n} - \frac{1}{n+1}$ (having decomposed into partial fraction)

$$a_1 = \frac{1}{1} - \cancel{\frac{1}{2}}$$

$$a_2 = \cancel{\frac{1}{2}} - \cancel{\frac{1}{3}}$$

$$a_3 = \cancel{\frac{1}{3}} - \cancel{\frac{1}{4}}$$

$\vdots$

$$a_{n-1} = \cancel{\frac{1}{n-1}} - \cancel{\frac{1}{n}}$$

$$a_n = \cancel{\frac{1}{n}} - \frac{1}{n+1}$$

Adding all the n^{th} equation gives,

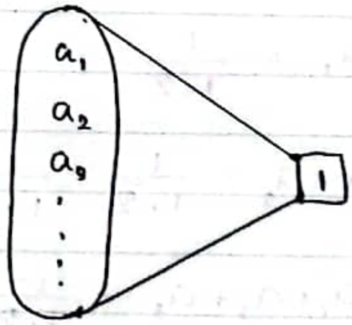
$$\sum_{k=1}^{n} a_k = 1 - \frac{1}{n+1} = S_n$$

$$\lim_{n \to \infty} S_n = 1$$

$\therefore$ The Series $\sum_{n=1}^{\infty} a_n$ is Convergent and has the Sum equal to 1

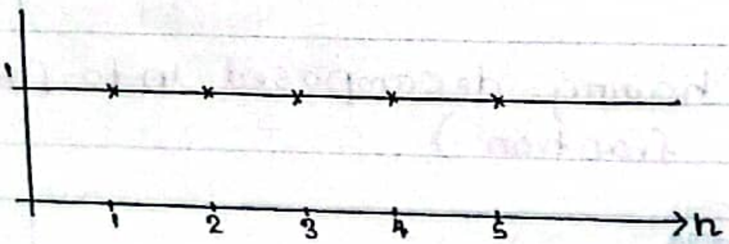
i.e;

$$\sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1$$



Ex 02 :- Consider $\sum_{n=1}^{\infty} 1$

Let $a_n = 1 \quad \forall n \in \mathbb{N}$



Then,

$S_n = a_1 + a_2 + a_3 + \ldots + a_n$

$= 1 + 1 + 1 + \ldots + 1$

$= n$

$\lim_{n \to \infty} S_n \Rightarrow \lim_{n \to \infty} n = \infty$

$\therefore (S_n)$ is divergent and so $\sum_{n=1}^{\infty} 1$ is divergent.

Ex 03 :- Consider $\sum_{n=1}^{\infty} 0$

Let $a_n = 0 \quad \forall n \in \mathbb{N}$

Then $S_n = a_1 + a_2 + \ldots + a_n$

$= 0 + 0 + \ldots + 0$

$= 0$

i.e; $S_n = 0$

$\therefore \lim_{n \to \infty} S_n = \lim_{n \to \infty} 0 = 0$

$\therefore (S_n)$ is Convergent and so, $\sum_{n=1}^{\infty} 0 = 0$

Ex 04 :- Consider $\sum_{n=1}^{\infty} 1/n$ Called a harmonic Series

Let $a_n = 1/n \quad \forall n \in \mathbb{N}$

$S_n = a_1 + a_2 + a_3 + \ldots + a_n$

$= 1 + 1/2 + 1/3 + \ldots + 1/n \quad \forall n \in \mathbb{N}$

Note that; the sequence (S_n) is not Cauchy.

Cauchy Sequence;

A sequence (a_n) is Cauchy if for any $\varepsilon > 0$ $\exists$ an N such that

$|a_n - a_m| < \varepsilon$ where $m, n > N$ Here $m, n \in \mathbb{N}$

Claim;

Take any two positive integers m and n with $n > m$

Now,

$$|S_n - S_m| = \left|\left(1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}\right) - \left(1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{m}\right)\right|$$

$$= \left|\frac{1}{m+1} + \frac{1}{m+2} + \dots + \frac{1}{n}\right|$$

$$= \frac{1}{m+1} + \frac{1}{m+2} + \dots + \frac{1}{n}$$

$$\geqslant \frac{1}{n} + \frac{1}{n} + \dots + \frac{1}{n}$$

$$= (n-m)\frac{1}{n}$$

$$= 1 - \frac{m}{n}$$

$n > m$

$\frac{1}{n} < \frac{1}{m}$

$m+1 < n$

$m+2 < n$

$m+3 < n$

$\frac{1}{m+1} > \frac{1}{n}$

$\frac{1}{m+2} > \frac{1}{n}$

i.e $|S_n - S_m| \geqslant 1 - \frac{m}{n}$

$\forall\, n, m \in \mathbb{N}$ with $n > m$

As $n > m$

Let $n = 2m$

Then $|S_n - S_m| > 1 - \frac{m}{n} = 1 - \frac{m}{2m} = \frac{1}{2} > 0$

$\therefore (S_n)$ is not a Cauchy Sequence //

Remark 01;

A Sequence of real numbers is Convergent. if and only if (iff) it is Cauchy.

Here, (S_n) is not Convergent. $\therefore \sum_{n=1}^{\infty} \frac{1}{n}$ is divergent

Also,

$$|S_{n+1} - S_n| = (1 + 1/2 + 1/3 + \ldots + 1/n+1) - (1 + 1/2 + 1/3 + \ldots + 1/n)$$

$$= 1/n+1 > 0 \quad \forall n \in \mathbb{N}$$

$\therefore (S_n)$ is monotonic increasing and it is not bounde above.

That is (S_n) is increasing and divergent.

$$\therefore \sum_{n=1}^{\infty} \frac{1}{n} = \infty$$

Theorem 01;

If $\sum_{n=1}^{\infty} a_n$ is Convergent then $\lim_{n\to\infty} a_n = 0$

$$\sum_{n=1}^{\infty} a_n \xrightarrow{\text{Convergent}} \lim_{n\to\infty} a_n = 0$$

Note:- This theorem says that $\sum_{n=1}^{\infty} a_n$ is Converger $\lim_{n\to\infty} a_n = 0$

Is the Converce of the theorem true?

If the $\lim_{n\to\infty} a_n = 0$

Then $\sum_{n=1}^{\infty} a_n$ is Convergent???

No!

Counter Example :-

Consider $\sum_{n=1}^{\infty} \frac{1}{n}$

Let $a_n = \frac{1}{n} \quad \forall n \in \mathbb{N}$

$\lim_{n\to\infty} a_n = \lim_{n\to\infty} \frac{1}{n} = 0$

But $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent.

Proof of theorem 1 ;

Let $S_n = \underbrace{a_1 + a_2 + a_3 + \dots + a_{n-1}}_{S_{n-1}} + a_n$

$$S_n = S_{n-1} + a_n$$
$$S_n - S_{n-1} = a_n$$

That is ; ; Let $\lim_{n\to\infty} S_n = S$

$$a_n = S_n - S_{n-1}$$

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} (S_n - S_{n-1})$$
$$= \lim_{n\to\infty} S_n - \lim_{n\to\infty} S_{n-1}$$
$$= S - S$$
$$= 0$$

Since $\sum_{n=1}^{\infty} a_n$ is Convergent (S_n) is Convergent.

<u>Test for divergence</u> ;

If $\lim_{n\to\infty} a_n \neq 0$, then $\sum_{n=1}^{\infty} a_n$ is divergent.

Ex :- Consider this $\sum_{n=1}^{\infty} n \sin(1/n)$

Let $a_n = n \sin(1/n) \quad \forall n \in \mathbb{N}$

$$\lim_{n \to \alpha} a_n = \lim_{n \to \alpha} n \sin(1/n)$$

$$= \lim_{n \to \alpha} \frac{\sin(1/n)}{1/n}$$

$\lim_{x \to 0} \frac{\sin x}{x} = 1$; where $x = 1/n$

$$\therefore \lim_{n \to \alpha} \frac{\sin(1/n)}{1/n}$$

$$= 1 \neq 0$$

$\therefore$ The Series $\sum_{n=1}^{\alpha} n \sin(1/n)$ is divergent.//

$\sum_{n=1}^{\alpha} \left(1 + \frac{1}{n}\right) \Rightarrow$ divergent.

Ex:- Consider this $\sum_{n=1}^{\alpha} (-1)^n$

Let $a_n = (-1)^n \ \forall n \in N$

$\lim_{n \to \alpha} a_n = \lim_{n \to \alpha} (-1)^n$ = does not exist

$$\therefore \lim_{n \to \alpha} a_n \neq 0$$

$\therefore$ $\sum_{n=1}^{\alpha} (-1)^n$ is divergent//.

* Alternative method;

$$S_n = a_1 + a_2 + a_3 + \dots + a_n = \begin{cases} 0 & \text{if } n \text{ is even} \\ -1 & \text{if } n \text{ is odd} \end{cases}$$

$\lim_{n \to \alpha} S_n$ DNE $\quad \sum_{n=1}^{\alpha} (-1)^n$ is divergent.

EX :-

Consider the Geometric Series with Common ratio r

$$\sum_{n=1}^{\infty} r^{n-1}$$

Let $a_k = r^{k-1}$, where $k = 1, 2, 3, \ldots\ldots$

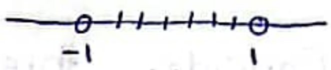
$$S_n = a_1 + a_2 + a_3 + \ldots\ldots + a_n$$

$$= 1 + r + r^2 + \ldots\ldots + r^{n-1}$$

$$= \begin{cases} \dfrac{1-r^n}{1-r} & \text{if } r \neq 1 \\ n & \text{if } r = 1 \end{cases}$$

Case i : $|r| < 1 \sim -1 < r < 1 \sim r \in (-1, 1)$

$$|r| = \begin{cases} r & \text{if } r \geqslant 0 \\ -r & \text{if } r < 0 \end{cases}$$



$$\lim_{n\to\infty} S_n = \lim_{n\to\infty} \frac{1-r^n}{1-r} = \frac{1}{1-r}$$

$\therefore$ The geometric Series is Convergent

i.e $\displaystyle\sum_{n=1}^{\infty} r^{n-1} = \frac{1}{1-r}$ if $|r| < 1$

Case ii :- $|r| \geqslant 1$

Let $a_n = r^{n-1}$

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} r^{n-1} \neq 0$$

$\therefore \displaystyle\sum_{n=1}^{\infty} a_n$ is not Convergent.

Ex :-

Consider $x = 0.999\ldots\ldots$

$$x = 0\times10^{0} + 9\times10^{-1} + 9\times10^{-2} + 9\times10^{-3} + \ldots\ldots$$

$$x = \frac{9}{10} + \frac{9}{10^2} + \frac{9}{10^3} + \ldots\ldots$$

$$= \frac{9}{10}\underbrace{\left[1 + \frac{1}{10} + \frac{1}{10^2} + \ldots\right]}_{\text{Geometric Series with } r = 1/10 < 1}$$

$$= \frac{9}{10}\left(\frac{1/10}{1 - 1/10}\right)$$

$$= \frac{9}{10} \times \frac{\cancel{10}}{\cancel{9}} \quad \frac{1}{9}$$

$$= 1$$

$$a + ar + ar^2 + \ldots + ar^{n-1} = \frac{a(1-r^n)}{1-r} \quad \text{if } r \neq 1$$

$$a + ar + ar^2 + \ldots = \lim_{n\to\infty} \frac{a(1-r^n)}{1-r}$$

$$= \frac{a}{1-r} \lim_{n\to\infty} (1-r^n)$$

$$= \frac{a}{1-r}$$

$$\lim_{n\to\infty} = 0$$

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r} \quad \text{if } r \neq 1$$

Theorem 02 :-

a) If the Series $\sum_{n=1}^{\alpha} a_n$ and $\sum_{n=1}^{\alpha} b_n$ Convertes, then $\sum_{n=1}^{\alpha} (a_n + b_n)$ Convertes and

$$\sum_{n=1}^{\alpha} (a_n + b_n) = \sum_{n=1}^{\alpha} a_n + \sum_{n=1}^{\alpha} b_n$$

b) If the Series $\sum_{n=1}^{\alpha} a_n$ Convertes and C is a constant then the Series $\sum_{n=1}^{\alpha} (C a_n)$ Converge and

$$\sum_{n=1}^{\alpha} (c\, a_n) = c \sum_{n=1}^{\alpha} a_n$$

Proof :-

Let S_n, t_n and u_n be the partial Sums of $\sum_{n=1}^{\alpha} a_n$, $\sum_{n=1}^{\alpha} b_n$ and $\sum_{n=1}^{\alpha} (a_n + b_n)$.

$$S_n = a_1 + a_2 + a_3 + \dots + a_n$$
$$t_n = b_1 + b_2 + b_3 + \dots + b_n$$
$$u_n = c_1 + c_2 + c_3 + \dots + c_n$$

where $C_k = a_k + b_k$; $k = 1, 2, 3, \dots, n$.

$$u_n = (a_1 + b_1) + (a_2 + b_2) + \dots + (a_n + b_n)$$
$$= (a_1 + a_2 + a_3 + \dots + a_n) + (b_1 + b_2 + b_3 + \dots + b_n)$$

we know that both $\sum_{n=1}^{\alpha} a_n$ and $\sum_{n=1}^{\alpha} b_n$ are Convergen

$\therefore$ The sequences (S_n) and (t_n) are Convergent.

$\therefore$ The Sequence $(S_n + t_n)$ is also Convergent.
Sums.

$$\therefore \lim_{n \to \alpha} (S_n + t_n) = \lim_{n \to \alpha} S_n + \lim_{n \to \alpha} t_n$$
$$= a + b$$ ~~where~~

mid 1

where ;

$a = \lim_{n\to\alpha} S_n$ and $\lim_{n\to\alpha} t_n$

$\therefore \lim_{n\to\alpha} u_n = a + b$

$\therefore \sum_{n=1}^{\alpha} (a_n + b_n)$ is convergent. and

$\sum_{n=1}^{\alpha} (a_n + b_n) = \sum_{n=1}^{\alpha} a_n + \sum_{n=1}^{\alpha} b_n$.

Ex :- Find $\sum_{n=1}^{\alpha} \frac{2^n - 4^{n+2}}{5^{n-1}}$

note:- $\frac{2^n - 4^{n+2}}{5^{n-1}} = 2\left(\frac{2}{5}\right)^{n-1} - 4^3\left(\frac{4}{5}\right)^{n-1}$

Both the Series $\sum_{n=1}^{\alpha}\left(\frac{2}{5}\right)^{n-1}$ and $\sum_{n=1}^{\alpha}\left(\frac{4}{5}\right)^{n-1}$ are geometric Series with ratios $r = 2/5$ and $r = 4/5$; < 1

Since their Common ratios are in $(-1, 1)$ then are convergent

$\sum_{n=1}^{\alpha}\left(\frac{2}{5}\right)^{n-1} = \frac{1}{1 - 2/5} = 5/3$ and $\sum_{n=1}^{\alpha}\left(4/5\right)^{n-1} = \frac{1}{1 - 4/5} = 5$

$\therefore \sum_{n=1}^{\alpha} \frac{2^n - 4^{n+2}}{5^{n-1}}$

$= 2 \sum_{n=1}^{\alpha}\left(2/5\right)^{n-1} - 4^3 \sum_{n=1}^{\alpha}\left(4/5\right)^{n-1}$

$= 2 \times 5/3 - 4^3 \times 5$

$= 10/3 - 320$

$= \frac{-950}{3}$

(x_n) is bounded if $\exists$ ~~x_n~~ an $M > 0$ such that $|x_n| < M$.

Theorem 03 :-

Let $\sum_{n=1}^{\infty} a_n$ be ~~x_n~~ an infinite series of nonnegative terms, $a_n \geq 0 \ \forall n \in \mathbb{N}$.

Then $\sum_{n=1}^{\infty} a_n$ converges if and only if the sequence of partial sums is bounded

i.e Let $S_n = \sum_{k=1}^{n} a_k$ $\sum_{n=1}^{\infty} a_n$ is Convergent iff $|S_n|$ is bounded.

Proof :-

Let $S_n = a_1 + a_2 + a_3 + \dots + a_n$ $\sum_{k=1}^{n} a_k$ — ① $\forall n \in \mathbb{N}$

① $\Rightarrow$ $S_{n+1} = a_1 + a_2 + a_3 + \dots + a_n + a_{n+1}$

$S_{n+1} = S_n + a_{n+1}$

$S_{n+1} - S_n = a_{n+1} \geq 0 \quad \forall n \in \mathbb{N}$

$\therefore S_{n+1} - S_n \geq 0$

That is, $S_{n+1} \geq S_n \ \forall n \in \mathbb{N}$

$S_1 \leq S_2 \leq S_3 \dots$

So, the sequence (S_n) is monotonic increasing

By the Monotonic Convergent theorem (MCT)

The sequence (S_n) is Convergent iff (S_n) is bounded.

i.e $\sum_{n=1}^{\infty} a_n$ is Convergent iff (S_n) is bounded (bdd)

Theorem 04 :-

The infinite Series $\sum_{n=1}^{\infty} a_n$ Convertes iff for each $\varepsilon > 0$ $\exists$ a Positive integer $N(\varepsilon)$ Such that ;

$$|a_{n+1} + a_{n+2} + \dots + a_m| < \varepsilon \quad \text{whenever } m > n > N(\varepsilon$$

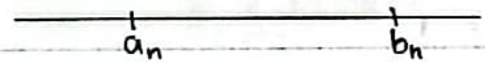
Theorem 05 :- (Comparison's test)

Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ be infinite Series of non-negative terms a_n and b_n with $a_n \leq b_n$

a) If $\sum_{n=1}^{\infty} b_n$ Converges , then $\sum_{n=1}^{\infty} a_n$ Converges

b) If $\sum_{n=1}^{\infty} a_n$ diverges , then $\sum_{n=1}^{\infty} b_n$ diverges .

$a_n, b_n \geq 0$ and $a_n \leq b_n$



$\sum a_n$		$\sum b_n$
cgt	$\longleftarrow$	cgt
dvg	$\longrightarrow$	dvg .

Ex :- Show that $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is Convergent.

Let $a_n = 1/n^2 \quad \forall n \in \mathbb{N}$

$$a_n = \frac{1}{n^2} = \left(\frac{1}{n}\right)\left(\frac{1}{n}\right) < \frac{1}{n} \frac{1}{(n-1)}$$

$$= \frac{1}{n(n-1)} \qquad \forall n = 2, 3, 4, \dots$$

Let $b_n = \dfrac{1}{n(n-1)} \quad \forall n \geq 2.$

$$b_n = \frac{-1}{n} + \frac{1}{n-1}$$

$$b_2 = -\tfrac{1}{2} + \tfrac{1}{1}$$

$$b_3 = -\tfrac{1}{3} + \tfrac{1}{2}$$

$$\vdots$$

$$b_{n-1} = \frac{-1}{n-1} + \tfrac{1}{n}$$

$$b_n = -\tfrac{1}{n} + \frac{1}{n-1}$$

$+$

$$\therefore \sum_{k=2}^{n} b_k = 1 - \frac{1}{k}$$

$$\lim_{n \to \alpha} \sum_{k=2}^{n} b_k = \lim_{k \to \alpha} \left(1 - \frac{1}{k}\right)$$

$$\Rightarrow \sum_{n=2}^{\alpha} b_n = 1$$

$$\therefore \sum_{n=1}^{\alpha} b_n = b_1 + \sum_{n=2}^{\alpha} b_n$$

$$= b_1 + 1$$

$\therefore \sum_{n=1}^{\alpha} b_n$ is Convergent.

$\therefore$ By Comparisan's test,

$\sum_{n=1}^{\alpha} a_n = \sum_{n=1}^{\alpha} \frac{1}{n^2}$ is Convergent.

CT : Comparison's Test .

Ex :-
Show that $\sum_{n=1}^{\infty} \frac{2^n}{3^n (n+1)^2+9}$ is Convergent.

Let $a_n = \frac{2^n}{3^n (n+1)^2+9} > 0 \quad \forall n \in \mathbb{N}$

$$a_n = \frac{2^n}{3^n (n+1)^2+9} > b_n$$

$$a_n = \frac{2^n}{3^n (n+1)^2+9} < \frac{2^n}{3^n} = \left(\frac{2}{3}\right)^n \quad \forall n \in \mathbb{N}$$

Let $b_n = \left(\frac{2}{3}\right)^n$

$$\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^n = \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)\left(\frac{2}{3}\right)^{n-1}$$

$$= \frac{2}{3} \underbrace{\sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^{n-1}}$$

A geometric Series
with Common ratio $r = \frac{2}{3} < 1$

$\therefore$ It is Convergent.
Now by the CT, $\sum_{n=1}^{\infty} a_n$ is Convergent.

ex :- Consider this $\sum_{n=1}^{\infty} \frac{1}{5n-3}$

Let $c_n = \frac{1}{5n-3} > 0 \quad \forall n \in \mathbb{N}$

Note that $\frac{1}{5n-3} > \frac{1}{5n} \quad \forall n \in \mathbb{N}$

Let $a_n = \frac{1}{5n}$ and $b_n = \frac{1}{5n-3} \quad \forall n \in \mathbb{N}$ $\sum \frac{1}{5n}$

Since $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent, $\sum_{n=1}^{\infty} a_n$ is divergent.

$\therefore \sum_{n=1}^{\infty} b_n$ is divergent.//.

ex :- Consider $\sum_{n=1}^{\infty} \frac{1}{2n-15}$

$2n - 15 > 0$

$n > 7.5$

$a_n > 0$ if $n \geq 8$.

$$\therefore \sum_{n=1}^{\infty} a_n = (a_1 + a_2 + a_3 + a_4 + a_5 + a_6 + a_7) + \sum_{n=8}^{\infty} \frac{1}{2n-15}$$

Let $a_n = \frac{1}{2n-15}$

$a_1 = -\frac{1}{13} < 0$	$a_5 = -\frac{1}{5} < 0$
$a_2 = \frac{-1}{11} < 0$	$a_6 = -\frac{1}{3} < 0$
$a_3 = -\frac{1}{9} < 0$	$a_7 = -\frac{1}{1} < 0$
$a_4 = -\frac{1}{7} < 0$	$a_8 = \frac{1}{1} > 0$.

Note that $\frac{1}{2n-15} > 0 \quad \forall\ \cancel{n \in \mathbb{N}}\ n \geq 8$.

$\frac{1}{2n-15} > \frac{1}{2n} \quad \forall n \geq 8$.

Then $a_n > a_n'$

Let $a_n' = \frac{1}{2n}$

Since $\sum_{n=1}^{\infty} a_n'$ is divergent. by CT,

$\sum_{n=0}^{\infty} a_n$ is divergent.

$\therefore \sum_{n=1}^{\infty} a_n$ is divergent.

Definition :-

If $\sum_{n=1}^{\infty} |a_n|$ Converges, then the series $\sum_{n=1}^{\infty} a_n$ is Said to be absolutely Convergent.

Ex :-

Consider $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$

Let $a_n = \frac{(-1)^n}{n^2} \quad \forall n \in \mathbb{N}$

$\Rightarrow |a_n| = \left|\frac{(-1)^n}{n^2}\right| = \frac{1}{n^2}\left|(-1)^n\right| = \frac{1}{n^2}$

we know that the Series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is Convergent.

$\therefore$ The Series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$ is absolutely Convergent.

Theorem 6 :-

If a Series is absolutely Convergent, Then it is Convergent.

$$\sum_{n=1}^{\infty} |a_n| \xRightarrow{cgt} \sum_{n=1}^{\infty} a_n \text{ cgt.}$$

$$\sum_{n=1}^{\infty} a_n \overset{cgt}{\not\Rightarrow} \sum_{n=1}^{\infty} |a_n| \text{ cgt.}$$

Ex :-

Consider $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$

we shall later Show that the Series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ convergent.

Let $a_n = \frac{(-1)^n}{n}$

Then $|a_n| = \left|\frac{(-1)^n}{n}\right| = 1/n$

Also we know that Series $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent.
i.e; the Series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ is Convergent. but
Not it is absolutely Convergent.

Theorem 07 :- (Limit form of CT)

Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ be infinite Series

of Positive terms: (i.e, $a_n, b_n > 0$).
Such that; ~~limit~~

$$\lim_{n\to\infty} \frac{a_n}{b_n} = l.$$

if $0 < l < \infty$, then two Series Converge or diverge together

$\sum a_n$		$\sum b_n$
cgt	$\Longrightarrow$	cgt
cgt	$\Longleftarrow$	cgt
dvg	$\Longrightarrow$	dvg.
dvg	$\Longleftarrow$	dvg.

Ex :-

Consider $\sum_{n=1}^{\infty} \frac{n}{8n^3+9}$

Let $a_n = \frac{n}{8n^3+9} > 0 \quad \forall n \in \mathbb{N}$

Note :-

$$\frac{1}{n^2} > \frac{n}{8n^3} > \frac{n}{8n^3+9}$$

Let $b_n = \frac{1}{n^2} > 0 \quad \forall n \in \mathbb{N}$

$$l = \lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \left[\frac{\frac{n}{8n^3+9}}{1/n^2} \right]$$

$$= \lim_{n\to\infty} \frac{n^3}{8n^3+9}$$

$$= \lim_{n\to\infty} \frac{1}{8+9/n^3}$$

$$= 1/8 > 0$$

Also, the Series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is Convergent.

$\therefore$ by CT, the Series $\sum_{n=1}^{\infty} \frac{n}{8n^3+9}$ is Convergent.

Ex:- Consider $\sum_{n=1}^{\infty} \frac{n}{8n^3-9}$

$a_n > 0$ iff $8n^3 - 9 > 0$

$$8n^3 > 9$$

$$n > \frac{9^{1/3}}{2}.$$

$\therefore a_n > 0 \quad \forall n \geq 2.$

Let $a_n = \frac{n}{8n^3-9}$

$a_1 = \frac{1}{8-9} = -1$

$$\sum_{n=1}^{\infty} a_n = a_1 + \sum_{n=2}^{\infty} a_n$$

b_n a_n.

$$\left(\frac{n}{8n^3-9}\right) > \frac{n}{8n^3} = \left(\frac{1}{8n^2}\right) = a_n' \text{ (Say)}$$

$$l = \lim_{n\to\infty} \frac{a_n}{b_n} = \lim_{n\to\infty} \frac{1/8n^2}{\frac{n}{8n^3-9}}$$

$$= \lim_{n\to\infty} \frac{8n^3-9}{8n^3}$$

$$= \lim_{n\to\infty} \frac{8 - 9/n^3}{8}$$

$$= 1 > 0 \quad (*)$$

But we know that $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is Convergent (*),

$\therefore$ By CT, the given Series is Convergent.

Determine the following Series are Convergent or divergent,

i) $\sum_{n=1}^{\infty} \sin(1/n)$

(ii) $\sum_{n=1}^{\infty} \frac{\sin(n\theta)}{n^2}$, where $\theta \in \mathbb{R}$.

Theorem 08 :- (Integral test).

Let f be a non negative decreasing continuous function defined a_n $[1, \alpha)$. ~~Thus~~ Then the $\sum_{n=1}^{\alpha} f_n$ Converges iff.

$\lim_{n \to \alpha} \int_1^n f(x)\, dx$ exists as a real number

i) $f(x) \geqslant 0 \quad \forall\, x \in [1, \alpha)$

$\sum_{n=1}^{\alpha} \frac{1}{n^2+2n}$ $\boxed{f_n = f(n)}$

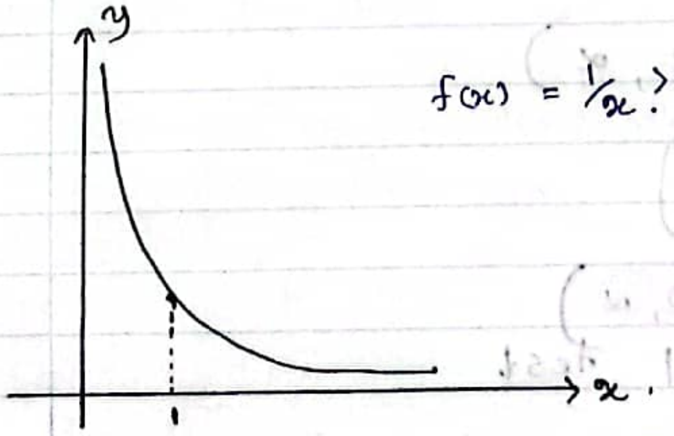
ii) f is $\searrow$ on $[1, \alpha)$

$f_n = \frac{1}{n^2+2n}$

(iii) f is Continuous on $[1, \alpha)$ $f(x) = \frac{1}{x^2+2x}$

Ex :- use the integral test to show that $\sum_{n=1}^{\alpha} \frac{1}{n}$ is divergent.

So, Let $f(x) = \frac{1}{x}$ for $x > 0$



(i) $f(x) > 0 \quad \forall\, n \in [1, \alpha)$

(ii) $f(x)$ is decreasing on $[1, \alpha)$

(iii) $f(x)$ is Continuous on $[1, \alpha)$

$\therefore$ we may apply the integral test

$$\lim_{n\to\alpha} \int_1^n f(x)\,dx = \lim_{n\to\alpha} \left[\int_1^n \frac{1}{x}\,dx \right]$$

$$= \lim_{n\to\alpha} \Big[\ln |x|\Big]_{x=1}^{n}$$

$$= \lim_{n\to\alpha} \Big[\ln n - \ln 1\Big]$$

$$= \alpha$$

Thus, by the integral test the given Series is divergent.

Ex:- Consider $\sum_{n=2}^{\alpha} \frac{1}{n\ln(n)}$

Let $f(x) = \frac{1}{x\ln(x)}$, $x > 1$

Then

i) $f(x) > 0 \quad \forall n \in [2, \alpha)$

(ii) $f(x) \searrow$ on $[2, \alpha]$

(iii) f is C+S on $[2, \alpha)$

Now by the integral test.

$$\lim_{n\to\alpha} \int_2^n \frac{1}{x\ln(x)}\,dx = \lim_{n\to\alpha} \int_2^n \frac{1}{\ln(x)}\,d[\ln(x)]$$

$$= \lim_{n\to\alpha} \Big[\ln(\ln(x))\Big]_{x=2}^{\alpha}$$

$$= \lim_{n \to \alpha} \left[\ln[\ln(x)] - \ln[\ln(2)] \right]$$

$$= \alpha$$

$\therefore$ By the integral the given Series is divergent.

HW.

(1) Consider the Series $\sum_{n=1}^{\alpha} \frac{1}{n^p}$ where, $p \in \mathbb{R}$

This is Called P - Series, use the Integral test to determine the diverges or Convergent

i) Case 1 ; $p < 0$ dvt

ii) Case 2 ; $p = 0$ dvt

iii) Case 3 ; $0 < p < 1$ dvt

iv) Case 4 ; $p = 1$ dvt

v) Case 5 ; $p > 1$ cgt.

(2) use P - Series test to determine the Converge of the following

(i) $\sum_{n=1}^{\alpha} \frac{1}{\sqrt{n}}$ (dvt)

(ii) $\sum_{n=1}^{\alpha} \frac{1}{n^{3/2}}$

3) Show that the Series $\sum_{n=1}^{\alpha} \frac{(-1)^n (n^2+3)}{n^6 + 5n^3 + 7}$ is Converg

(use P - Series + LCT). Limit form of.

4) use the Integral test to determine the Converges / divergence. of the Series. $\sum_{n=2}^{\alpha} \frac{1}{n \ln(n) \ln(\ln(n))}$

Theorem 09 (Ratio test).

Let $\sum a_n$ be a Series with non-Zero terms.

(a) If there exist a positive number $r<1$ and a Positive integer, $N \in \mathbb{N}$ Such that

$$\left|\frac{a_{n+1}}{a_n}\right| \leqslant r < 1 \quad \forall\ n \geqslant N$$

Then $\sum a_n$ is absolutely Convergent. (i.e., It is Convergent)

(b) if there exist a number $r \geqslant 1$ and a positive integer $N \in \mathbb{N}$ Such that

$$\left|\frac{a_{n+1}}{a_n}\right| \geqslant r > 1 \quad n > N,$$ then $\sum a_n$ is divergent.

Theorem 10. (Limit form of ratio test).

Let $\sum_{n=1}^{\infty} a_n$ be a Series with non-Zero terms

Let ~~limit~~ $\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \ell$.

(a) if $\ell < 1$, then $\sum_{n=1}^{\infty} a_n$ is absolutely Conver(gent) (that is Convergent)

(b) if $\ell > 1$, then $\sum_{n=1}^{\infty} a_n$ is divergent.

(c) if $l = 1$, then the test is inconclusive,

Ex:- Consider $\sum_{n=1}^{\infty} \frac{(-1)^{n-1} 3^n}{\lfloor n}$

Let $a_n = \frac{(-1)^{n-1} 3^n}{\lfloor n} \quad \forall n \in \mathbb{N}$

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{(-1)^n 3^{n+1}}{\lfloor n+1} \times \frac{\lfloor n}{(-1)^{n-1} 3^n}\right|$$

$$= \left|\frac{-3}{n+1}\right| = \frac{3}{n+1}$$

$$\therefore l = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$$

$$= \lim_{n\to\infty} \frac{3}{n+1}$$

$$= 0 < 1$$

$\therefore$ from the limit form of the ratio test the given Series absolutely cgt and hence it is convergent.

Ex:- Consider $\sum_{n=1}^{\infty} \frac{1.3.5\ldots(2n-1)}{\lfloor n}$

Let $a_n = \frac{1.3.5\ldots(2n-1)}{\lfloor n} \quad \forall n \in \mathbb{N}$

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{1.3.5\ldots(2n-1)(2n+1)}{\lfloor (n+1)} \times \frac{\lfloor n}{1.3.5\ldots(2n-1)}\right|$$

$$= \frac{2n+1}{n+1} = \frac{2+1/n}{1+1/n}$$

$$\ell = \lim_{n\to\alpha} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\alpha} \frac{2+1/n}{1+1/n} = 2 > 1$$

$\therefore$ the Series is dvg (according to the lim form of ratio test).

eg:- Consider $\sum_{n=1}^{\alpha} (-1)^n$

Let $a_n = (-1)^n \quad \forall n \in \mathbb{N}$

$$\left|\frac{a_{n+1}}{a_n}\right| = \left|\frac{(-1)^{n+1}}{(-1)^n}\right| = |-1| = 1$$

$$\therefore \lim_{n\to\alpha} \left|\frac{a_{n+1}}{a_n}\right| = \ell.$$

$$\ell = \lim_{n\to\alpha} 1$$

$$= 1$$

$\therefore$ limit form of ratio test Cannot be applie

But we know that $\sum_{n=1}^{\alpha} (-1)^n$ is divergent.

eg:- Consider $\sum_{n=1}^{\alpha} \frac{1}{n^2}$

Let $a_n = 1/n^2 \quad \forall n \in \mathbb{N}$

$$\frac{a_{n+1}}{a_n} = \left|\frac{1}{(n+1)^2} \times n^2\right|$$

$$= \frac{n^2}{(n+1)^2}$$

$$= \left(\frac{n}{n+1}\right)^2 = \left(\frac{1}{1+1/n}\right)^2$$

$$\therefore \ l = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left[\frac{1}{1+1/n}\right]^2$$

$$= 1$$

$\therefore$ the limit form of the ratio test can not be applied. But the series $\sum_{n=1}^{\infty} \frac{1}{n^2}$ is convergent.

Theorem 11 (Alternative Series test).

Let C_n be a decreasing sequence of positive numbers. Also suppose that $\lim_{n\to\infty} C_n = 0$ then the series $\sum_{n=1}^{\infty} (-1)^{n-1} C_n$ and $\sum_{n=1}^{\infty} (-1)^n C_n$ converge.

* C_n is $\downarrow$ $C_{n+1} - C_n \leq 0 \quad \forall n \in \mathbb{N}$.
* $C_n > 0 \quad \forall n \in \mathbb{N}$
* $\lim_{n\to\infty} C_n = 0$

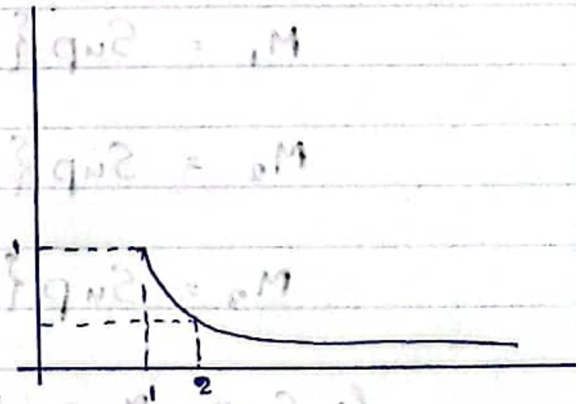
Eg:- Consider $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$

Let $C_n = \frac{1}{n}$. $\quad C_n > 0 \ \forall n \in \mathbb{N}$

$$C_{n+1} - C_n = \frac{1}{n+1} - \frac{1}{n}$$

$$= \frac{n - n - 1}{n(n+1)}$$

$$= \frac{-1}{n(n+1)} < 0$$



$\therefore C_n$ is a decreasing sequence.

$$\lim_{n\to\infty} c_n = \lim_{n\to\infty} \frac{1}{n} = 0.$$

$\therefore$ we can apply alternative series test. Hence the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ is Convergent.

Root test.

Def^n :- Let (x_n) be a Seq^n of real numbers, t Lim Sup x_n and Lim inf x_n are define by,

$$\lim\sup x_n = \inf\left(\sup_{k\geq n} x_k\right)$$

$$= \inf\left\{\sup\{x_n, x_{n+1}, \dots\}\right\}$$ an

$$\lim\inf x_n = \sup_{n\geq 1}\left(\inf_{k\geq n} x_k\right)$$

$$= \sup\left\{\inf\{x_n, x_{n+1}, \dots\}\right\}$$

respectively.

Let $M_n = \sup\{x_n, x_{n+1}, x_{n+2}, \dots\}$

$M_1 = \sup\{x_1, x_2, x_3, \dots\}$

$M_2 = \sup\{x_2, x_3, x_4, \dots\}$

$M_3 = \sup\{x_3, x_4, x_5, \dots\}$

$\lim\sup x_n = \inf\{M_1, M_2, M_3, \dots\}$

Also Let $m_n = \inf\{x_n, x_{n+1}, x_{n+2}, \dots\}$

$m_1 = \inf\{x_1, x_2, x_3 \ldots\}$.

$m_2 = \inf\{x_2, x_3, x_4, \ldots\}$.

$m_3 = \inf\{x_3, x_4, x_5, \ldots\}$

$\lim\inf x_n = \text{Sup}\{m_1, m_2, m_3, \ldots\}$.

For example, Let $x_n = 1/n$

$M_n = \text{Sup}\{x_n, x_{n+1}, x_{n+2}, \ldots\}$

$M_1 = \text{Sup}\{1, 1/2, 1/3, 1/4, \ldots\} = 1$

$M_2 = \text{Sup}\{1/2, 1/3, 1/4, 1/5, \ldots\} = 1/2$

$M_3 = \text{Sup}\{1/3, 1/4, 1/5, 1/6, \ldots\} = 1/3$

$\lim \text{Sup}\, x_n = \inf\{M_1, M_2, M_3, \ldots\}$

$= \inf\{1, 1/2, 1/3, \ldots\}$.

$= 0$.

Let $m_n = \inf\{x_n, x_{n+1}, x_{n+2}, \ldots\}$

$m_1 = \inf\{1, 1/2, 1/3, \ldots\} = 0$.

$m_2 = \inf\{1/2, 1/3, 1/4, \ldots\} = 0$

$m_3 = \inf\{1/3, 1/4, 1/5, \ldots\} = 0$.

$\therefore \liminf x_n = \operatorname{Sup}\{m_1, m_2, m_3, \ldots\}$.

$= \operatorname{Sup}\{0, 0, 0, \ldots\}$.

$= 0$

Find $\limsup x_n$ and $\liminf x_n$ when $x_n = (-1)$ $\forall n \in \mathbb{N}$

$\limsup x_n = 1$
$\liminf x_n = -1$

Let (x_n) be a Sequence of real numbers.
Let $y_n = \operatorname{Sup}\{x_n, x_{n+1}, x_{n+2}, \ldots\}$.

$= \operatorname{Sup} x_k$
$= k \geq n$.

Then, $\limsup x_n = \lim_{n \to \alpha} y_n$ and

i.e. $\limsup x_n = \lim_{n \to \alpha} \left(\operatorname{Sup}_{k \geq n} x_k\right)$

Let $x_n = \inf x_k = \inf\{x_n, x_{n+1}, x_{n+2}, \ldots\}$.

$\liminf x_n = \lim_{n \to \alpha} \cancel{x_n}\ Z_n = \lim_{n \to \alpha} \left(\inf_{k \geq n} x_k\right)$

$x_n = 1/n \qquad y_n = \operatorname{Sup}\{1/n, \frac{1}{n+1}, \frac{1}{n+2}, \ldots\}$.

$= 1/n$

$$\lim\sup x_n = \lim_{n\to\alpha} 1/n = 0.$$

$$z_n = \inf\{1/n, 1/n+1, 1/n+2, \ldots\}.$$

$$= 0$$

$$\lim\inf x_n = \lim_{n\to\alpha} 0 = 0.$$

Theorem 12 - Root test.

Let $\sum a_n$ be a given series and let $\alpha = \lim\sup |a_n|^{1/n}$

a) If $\alpha < 1$, then $\sum a_n$ is absolutely convergent.
b) If $\alpha > 1$, then $\sum a_n$ is divergent.
c) If $\alpha = 1$, then the test fails.

Consider $\sum_{a=1}^{\alpha} (-1)^{n+1}\left(\frac{2n+7}{8n-2}\right)^n$

Let $a_n = (-1)^{n+1}\left(\frac{2n+7}{8n-2}\right)^n$

$$|a_n| = \left|(-1)^{n+1}\left(\frac{2n+7}{8n-2}\right)^n\right| = |(-1)^{n+1}|\left|\left(\frac{2n+7}{8n-2}\right)^n\right|$$

$$= \left(\frac{2n+7}{8n-2}\right)^n$$

$$\Rightarrow |a_n|^{1/n} = \frac{2n+7}{8n-2}$$

$$\alpha = \lim\sup|a_n|^{1/n} = \lim\sup\left(\frac{2n+7}{8n-2}\right) \quad — (1)$$

$$\operatorname{Sup}\left\{\frac{2n+7}{8n-2} \,\middle|\, n \in \mathbb{N}\right\} = \left\{\frac{9}{6}, \frac{11}{14}, \frac{13}{22}, \ldots\right\}$$

$$y_n = \operatorname{Sup}\{x_n, x_{n+1}, x_{n+2}, \ldots\}.$$

$$= \operatorname{Sup}\left\{\frac{2n+7}{8n-2}, \frac{2(n+1)+7}{8(n+1)-2}, \frac{2(n+2)+7}{8(n+2)-2}, \ldots\right.$$

$$= \frac{2n+7}{8n-2}.$$

From ①, $\alpha = \lim_{n\to\alpha} \frac{2n+7}{8n-2}$

$$= \lim_{n\to\alpha} \frac{2+7/n}{8-2/n}$$

$$= 2/8$$

$$= 1/4 < 1$$

$\therefore$ by the root test $\sum a_n$ is absolutely Convergent So, it is Convergent.

Ex:- Cosider $\sum_{n=1}^{\alpha} \left[\frac{3+(-1)^n}{3+2(-1)^n}\right]^n$

Let $a_n = \left[\frac{3+(-1)^n}{3+2(-1)^n}\right]^n \quad \forall\, n \in \mathbb{N}$

$$|a_n| = \left|\frac{3+(-1)^n}{3+2(-1)^n}\right|^n$$

$$|a_n|^{1/n} = \frac{3+(-1)^n}{3+2(-1)^n} = \begin{cases} 4/5 & \text{if } n \text{ is even} \\ 2 & \text{if } n \text{ is odd.} \end{cases}$$

$$\alpha = \limsup |a_n|^{1/n}$$
$$= \limsup \left[\frac{3+(-1)^n}{3+2(-1)^n}\right]$$
$$= 2 > 1$$

$\therefore$ By the root test, the series is divergent.

Ex:- Consider $\sum_{n=1}^{\alpha} \left[\frac{2+(-1)^n}{3+2(-1)^n}\right]^n$.

Let $a_n = \left[\frac{2+(-1)^n}{3+2(-1)^n}\right]^n \quad \forall n \in \mathbb{N}$

$$\Rightarrow |a_n|^{1/n} = \frac{2+(-1)^n}{3+2(-1)^n} = \begin{cases} 1 & \text{if } n \text{ is odd} \\ 3/5 & \text{if } n \text{ is even.} \end{cases}$$

$$\alpha = \limsup |a_n|^{1/n}.$$
$$= \limsup \frac{2+(-1)^n}{3+2(-1)^n}$$
$$= 1$$

$\therefore$ The Root test cannot be applied to determine whether the series is convergent or divergent.

*. Try using limit form of RT and RT - $\sum_{n=1}^{\alpha} \frac{n!}{n^n}$

$$\left[\lim_{n \to \alpha} \left(1+\frac{x}{n}\right)^n = e^x\right]$$

Chapter 02 - Power Series.

A Series of the form $a_0 + a_1(x-c) + a_2(x...$ $.... + a_n(x-c)^n + ... = \sum_{n=0}^{\infty} a_n (x-c)^n$ is Called a power Series.

Centered at c, where $a_n \in \mathbb{R}$ $\forall$ $n \in \mathbb{N}$ and $c \in \mathbb{R}$

Ex 01:- Consider the power series $\sum_{n=0}^{\infty} x^n$

$$\sum_{n=0}^{\infty} x^n = \sum_{n=0}^{\infty} 1 \times (x-0)^n$$

Here $a_n = 1$ $\forall$ $n \in \mathbb{N}$

$c = 0 \rightarrow$ the Power Series centered.

when $x = 1$;

The power series becomes $\sum_{n=0}^{\infty} 1 \cdot 1^n = \sum_{n=0}^{\infty} 1$ is divergent.

when $x = -1$;

The Power Series becomes $\sum_{n=0}^{\infty} 1 \cdot (-1)^n = \sum_{n=0}^{\infty} (-1)^n$ also divergent.

when $x = 0$;

The Power Series becomes $\sum_{n=0}^{\infty} 1 \times 0^n = \sum_{n=0}^{\infty} 0$ is Convergent.

when $x = 1/2$;

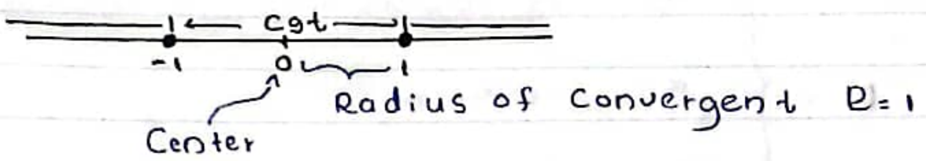
The Power Series becomes $\sum_{n=0}^{\infty} 1 \times x^n = \sum_{n=0}^{\infty} (1/2)$

(a geometric Series with Common ratio $r = 1/2 < 1$

Since $|r| = \frac{1}{2} < 1$ this is cgt. ($\sum x^n$ is cgt $\forall$ $|x| <$

Infact $\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \dots + x^n + \dots$ is a geomatric

Series with Common ratio x. Hence the power Series $\sum_{n=0}^{\infty} x^n$ is Convergent. when $|x| < 1$ and divergent. when $|x| \geq 1$



$(-1, 1)$ - interval of Convergence.

Ex 02 :- Consider $\sum_{n=0}^{\infty} (n+1)^2 (x-3)^2$.

Here $a_n = (n+1)^2$, $C = 3$

This is a Power Series Centered at $C = 3$

when $x = 3$;

The Power Series becomes $\sum_{n=0}^{\infty} 0$ is Convergent

when $x = 2$;

The Power Series becomes $\sum_{n=0}^{\infty} (n+1)^2 (-1)^n$ is not Convergent.

The Convergent of a Power Series depends both x and a_n.

Theorem 01 ;

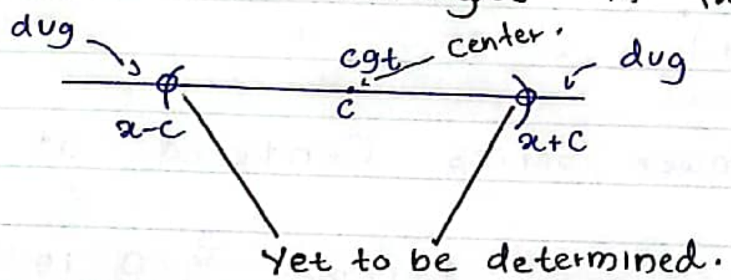
Let $\sum_{n=0}^{\infty} a_n (x-c)^n$ be a Power Series

Let $\lambda = \limsup |a_n|^{1/n}$

Define R by

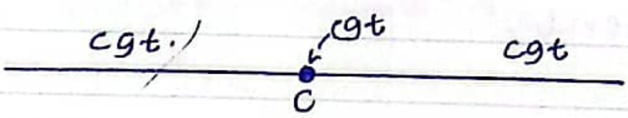
$$R = \begin{cases} 1/\lambda & \text{if } 0 < \lambda < \infty \\ \infty & \text{if } \lambda = 0 \\ 0 & \lambda = \infty \end{cases}$$

a) Suppose that $0 < R < \infty$.
Then the Power Series Converges absolutely if $|x-c| < R$ and diverges if $|x-c| > R$



R: The Radius of Convergence.

b) If $R = \infty$, then the Power Series Converges absolutely $\forall x$ in $\in (-\infty, \infty)$



The radius of Convergence $R = \infty$.
The interval of Convergence $= (-\infty, \infty)$

c) If $R = 0$, Then the Power Series Converges at $x = c$

$0 < R < \alpha \Rightarrow |x-c| < R \;\wedge\; |x-c| > R$
cgt dvg.

Proof :-

Clearly, the Power Series is absolutely Convergen at $x = c$,

So, Let $x \neq c$

Let $a_n' = a_n (x-c)^n$

Apply the root test to the Power Series.

$$\alpha = \limsup |a_n'|^{1/n}$$

$$= \limsup |a_n (x-c)^n|^{1/n}$$

$$= \limsup |a_n|^{1/n} (x-c)$$

$$= |x-c| \underbrace{\limsup |a_n|^{1/n}}_{\lambda}$$

$$= |x-c| \lambda \quad , \text{ since } \lambda = \limsup |a_n|^{1/n}.$$

Now by the root test,

$\sum_{n=0}^{\infty} a_n (x-c)^n$ converges absolutely if $\alpha < 1$ and diverges if $\alpha > 1$.

a) Suppose $0 < R < \alpha$

Now, $\alpha < 1 \Rightarrow (x-c).\lambda < 1$

$\Rightarrow (x-c) < 1/\lambda$

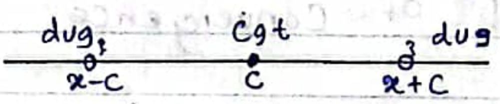
$\Rightarrow (x-c) < R$

i.e; The Power Series Converges absolutely when $(x-c) < R$.

when $\alpha > 1$ we have $|x-c| \lambda > 1$

$\Rightarrow |x-c| > 1/\lambda = R$

i.e; The Power Series diverges when $|x-c| > R$.

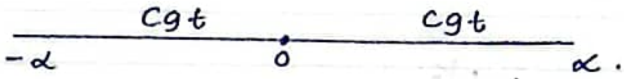


b) $R = \infty$

$\Rightarrow \lambda = 0$

$\therefore \alpha = |x-c|\lambda = |x-c| \times 0 = 0 < 1$

$\therefore$ The Power Series is $\sum_{n=0}^{\infty} a_n (x-c)^n$ Converges absolutely $\forall x \in \mathbb{R}$



c) Let $R = 0$

$\Rightarrow \lambda = \infty$

$\alpha = |x-c|\lambda = \infty$ Since $x \neq c$

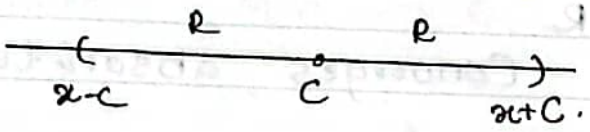
i.e; $\alpha > 1$

$\therefore$ The Power Series $\sum_{n=0}^{\infty} a_n (x-c)^n$ diverges $\forall x \neq c$

Ex 03:- Consider Power Series $\sum_{n=0}^{\infty} \frac{x^{\cancel{3}n}}{n}$

The interval of Convergence is an interval wher the given Power Series Converges.

The radius of Convergence $= R$.



$\sum_{n=0}^{\infty} \frac{x^n}{n} = \sum_{n=0}^{\infty} \frac{1}{n} (x-c)^n$

$a_n = 1/n \quad \forall n \in \mathbb{N}$

The Center of the Power Series $= 0$

we want to find the radius of Converges and the interval of Convergence

$\lambda = \limsup |a_n|^{1/n}$

$= \limsup \left|\frac{1}{n}\right|^{1/n} = \limsup \frac{1}{n^{1/n}}$

$= \lim_{n \to \infty} \frac{1}{n^{1/n}}$

$= 1$

$$\left\{ \lim_{n\to\infty} n^{1/n} = 1 \right\}$$

$R = \frac{1}{\lambda} = \frac{1}{1} = 1$

i.e ; the radius of Converges = 1

The Power Series Converges absolutely in $|x| < 1$

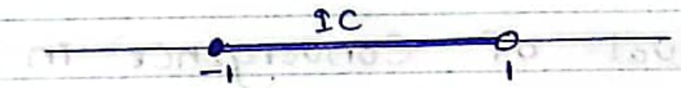
i.e ; The Power Series Converge absolutely in $(-1, 1)$

we have to check the Converges at $x = -1$ and $x = 1$

when $x = 1$, the Power Series becomes $\sum_{n=0}^{\infty} \frac{1}{n}$ (harmonic Series) This Series is divergent.

when $x = -1$, the Power Series becomes $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ which is Convergent (By AST)

∴ The interval of Convergence, $IC = [-1, 1)$



Ex:04 ; Find R and IC of the Power Series, $\sum_{n=1}^{\infty} \frac{x^n}{n^2}$

$$\sum_{n=1}^{\infty} \frac{x^n}{n^2} = \sum_{n=1}^{\infty} \frac{1}{n^2}(x-0)^n.$$

$a_n = 1/n^2 \quad \forall n \in \mathbb{N}$

The Center of the Power Series $(c) = 0$.

we want to find R and IC.

$$\lambda = \limsup |a_n|^{1/n}$$

$$= \limsup \left|\frac{1}{n^2}\right|^{1/n} = \limsup \frac{1}{n^{2/n}}$$

$$= \lim_{n\to\infty} \frac{1}{(n^{1/n})^2}$$

$$= 1$$

$R = 1/\lambda = 1/1 = 1$

$\therefore$ Radius of Convergent ; $R = 1$.

Then the power Series Converges absolutely if $|x| < 1$

That is, $(-1, 1)$

The Power Series in Converges $(-1, 1)$

we have to check Convergence of the Power Series $x = 1$ and $x = -1$

~~we~~ when $x = 1$, the Power Series becomes.

$\sum_{n=1}^{\infty} \frac{1}{n^2}$, which is Convergent.

when $x = -1$, the PS becomes,

$\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$, which is Convergent. (By AST)

$\therefore$ The interval of Convergence in $[-1, 1]$

g:-03. Find R and IC of $\sum_{n=0}^{\infty} \frac{x^n}{n^n}$

$$\sum_{n=0}^{\infty} \frac{x^n}{n^n} = \sum_{n=0}^{\infty} \frac{1}{n^n} (x-0)^n.$$

Here $a_n = 1/n^n$ and $c = 0$.

$$\lambda = \lim\sup |a_n|^{1/n}$$

$$= \lim\sup \left|\frac{1}{n^n}\right|^{1/n} = \lim\sup \frac{1}{n}$$

$$= \lim_{n \to \infty} \frac{1}{n}$$

$$= 0.$$

$R = 1/\lambda$

$R = \infty$

$\therefore$ The interval of convergence, $IC = (-\infty, \infty) \equiv \mathbb{R}$

Ex 06:- Find R and IC of $\sum_{n=1}^{\infty} n^n x^n$.

$$\sum_{n=1}^{\infty} n^n x^n = \sum_{n=1}^{\infty} n^n (x-0)^n.$$

Here; $a_n = n^n$ and $c = 0$.

$$\lambda = \lim\sup |a_n|^{1/n}$$
$$= \lim\sup |n^n|^{1/n} = \lim\sup n.$$

$$= \lim_{n\to\infty} n$$

$$= \infty$$

$R = 1/\lambda$

$R = 0.$

$\therefore$ Radius of Convergent ; $R = 0$

$\therefore$ The power series converges only at $x = 0$ //

H.W :-

1) Find R and IC of $\sum_{n=0}^{\infty} \left(\frac{2+3n}{7+5n}\right)^n x^n$.

Theorem 02 :-

Let $\sum_{n=0}^{\infty} a_n (x-c)^n$ be a Power Series with non-zero Coefficients,

If $\lim_{n\to\infty} \left|\frac{a_n}{a_{n+1}}\right|$ exist. (as a real ~~value~~ number or ∞

Then the radius of Convergence R is given by,

$$R = \lim_{n\to\infty} \left|\frac{a_n}{a_{n+1}}\right| \qquad \boxed{a_n \neq 0 \quad \forall n}$$

Proof :-

Let $a'_n = a_n (x-c)^n \quad \forall n \in \mathbb{N}$;

Apply the ratio test.

$$\left|\frac{a'_n}{a'_{n+1}}\right| = \left|\frac{a_n (x-c)^n}{a_{n+1}(x-c)^{n+1}}\right|$$

$$= \left|\frac{a_n}{a_{n+1}}\right| \frac{1}{|x-c|}$$

$$\therefore \left|\frac{a'_{n+1}}{a'_n}\right| = \left|\frac{a_{n+1}}{a_n}\right| |x-c| \text{ if } x \neq c$$

$$\therefore l = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| |x-c| = |x-c| \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$$

The power Series Converges absolutely if $l < 1$

i.e ; $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| |x-c| < 1$

i.e ; $|x-c| < \lim_{n\to\infty} \left|\frac{a_n}{a_{n+1}}\right|$

if $|x-c| < \lim_{n\to\infty} \left|\frac{a_n}{a_{n+1}}\right|$

The PS Converges and if $|x-c| > \lim_{n\to\infty} \left|\frac{a_n}{a_{n+1}}\right|$ the PS diverges.

~~The PS diverges.~~

$\therefore R = \lim_{n\to\infty} \left|\frac{a_n}{a_{n+1}}\right|$

$$\boxed{R = \lim_{n\to\infty} \left|\frac{a_n}{a_{n+1}}\right|}$$

Ex 07:- Find R and IC of $\sum_{n=0}^{\infty} \frac{n^3}{n!} x^n$

$$\sum_{n=0}^{\infty} \frac{n^3}{n!} x^n = \sum_{n=0}^{\infty} \frac{n^3}{n!} (x-c)^n.$$

Here $a_n = \frac{n^3}{n!}$ $\forall n \in \mathbb{N}$

$$R = \lim_{n\to\infty} \left|\frac{a_n}{a_{n+1}}\right| = \lim_{n\to\infty} \left|\frac{n^3}{n!} \times \frac{(n+1)!}{(n+1)^3}\right|$$

$$= \lim_{n\to\infty} \left|\frac{n^3}{\cancel{n!}} \times \frac{\cancel{(n+1)}\,\cancel{n!}}{(n+1)^{\cancel{3}2}}\right|$$

$$= \lim_{n\to\infty} \frac{n^3}{(n+1)^2}$$

$$= \lim_{n\to\infty} \frac{n}{[1+1/n]^2}$$

$R = \infty$

$\therefore$ The PS Converges $\forall x \in \mathbb{R}$

$\therefore$ The IC is $(-\infty, \infty)$ //.

H/W:- Find a R and IC of $\sum_{n=0}^{\infty} \left(\frac{7+5n}{4+3n}\right)^n x^{2n}$

Theorem 03 ;

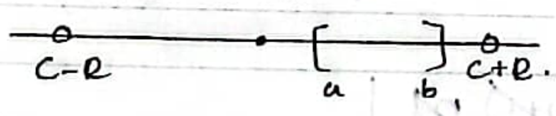
Let $R\ (>0)$ be the radius of Convergence of the Power Series $\sum_{n=0}^{\infty} a_n (x-c)^n$

Then ,

a) The Power Series is Continuous on the interval of Convergents.

b) The power Series Can be differentiated term term within the interval of Convergence. (IC)

c) The Power Series Can be integrated term by term over any interval $[a, b]$ of the $[a,b] \subseteq (c-R, c+R)$



Ex 08 :- Consider the Power Series $\sum_{n=0}^{\infty} x^n$ for $|x| < 1$

$$\sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \dots + x^n + \dots \quad \text{for } |x| < 1$$

IC is $(-1, 1)$

$\therefore$ 1) the Series is Continuous on IC

2) The Series Can be differentiated on IC

3) The Series Can be integrated term by term on IC

we know that.

$$\frac{1}{1-x} = 1 + x + x^2 + \dots ; \cancel{1+x^k} = \sum_{n=0}^{\infty} x^n \quad |x| < 1$$

Differentiation gives ,

$$-(1-x)^{-2} = 0 + 1 + 2x + 3x^2 + \dots ,$$

$$= \sum_{n=1}^{\infty} n x^{n-1} \quad \text{if } |x|<1$$

At $x=-1$ the PS becomes;

$\sum_{n=1}^{\infty} n(-1)^{n-1}$, which is not Convergent.

At $x=1$ the PS becomes.

$\sum_{n=1}^{\infty} n$, which is not Convergent.

$$\therefore \frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} n x^{n-1} \quad \forall\, x \in (-1,1)$$

Now according to theorem 03, we can the power Series term by term.

For $t \in (-1,1)$, we have;

$$\frac{1}{1-t} = \sum_{n=0}^{\infty} t^n$$

$$\int_0^x \frac{1}{1-t}\,dt = \int_0^x \left[\sum_{n=0}^{\infty} t^n\right] dt = \sum_{n=0}^{\infty} \int_0^x t^n\,dt \quad (\text{theorem } 03).$$

$$-\ln|1-t| \Big|_{t=0}^{x} = \sum_{n=1}^{\infty} \frac{t^{n+1}}{n+1} \Big|_{t=0}^{x}.$$

$$-\ln|1-x| = \sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1}$$

$$-\ln(1-x) = \sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1}, \quad \text{Since } \cancel{0<x<1}\ |x|<1$$

$$\ln(1-x) = -\sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1} \qquad |x|<1$$

At the end point $x=-1$, the Power Series becomes $-\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$ which is Convergent due to the AST.

So, -1 is a point of the interval of Convergence of the Power Series.

At the end point $x = +1 = 1$, the Power Series becomes ~~negative~~ $-\sum_{n=1}^{\infty} \frac{1}{n}$ which is divergent.

So, 1 is not a Point of IC.

$$\boxed{\ln(1-x) = -\sum_{n=1}^{\infty} \frac{x^n}{n} \quad \text{if} \quad -1 \leq x < 1}$$

Ex 09:- Find the power Series of $\frac{1}{1+x^2}$ for $|x|<1$

$$\frac{1}{1+x^2} = \sum_{n=0}^{\infty} C_n (x-c)^n$$

we know that ;

$$\frac{1}{1-t} = \sum_{n=0}^{\infty} t^n \quad \text{if} \quad |t| < 1 \quad \text{— ①}$$

$t = -x^2$, Provider $|x| < 1$

Now put $t = -x^2$ in ①.

Then we get,

$$\frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-x^2)^n \quad \text{if } |-x^2| < 1$$

$$= \sum_{n=0}^{\infty} (-1)^n x^{2n} \quad \text{if} \quad |x| < 1 \quad \text{— ②}.$$

Hence, find the PS of $\tan^{-1} x$ we know that for $|x| < 1$

② with x replaced by y gives

$$\frac{1}{1+y^2} = \sum_{n=1}^{\infty} (-1)^n y^{2n} \quad \text{for} \quad |y| < 1,$$

~~Provider $|x| < 1$~~

Intergrating gives ;

$$\int_0^x \frac{\cancel{\tan^{-1}(y)}\ 1\,dy}{1+y^2} = \int_0^x \left[\sum_{n=0}^{\infty} (-1)^n y^{2n}\right] dy = \sum_{n=0}^{\infty} \left[\int_0^x (-1)^n y^{2n}\,dy\right]$$

$$\tan^{-1} y \Big|_{y=0}^{x} = \sum_{n=0}^{\infty} (-1)^n \frac{y^{2n+1}}{2n+1} \Big|_{y=0}^{x}$$

$$\tan^{-1} x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} \quad \text{for } |x|<1$$

$\therefore \tan^{-1} x =$

At $x=-1$, the PS becomes

$$\sum_{n=0}^{\infty} \frac{(-1)^n \overbrace{(-1)^{2n+1}}^{\text{odd}}}{2n+1} = -\sum_{n=0}^{\infty} \frac{\cancel{8}(-1)^n}{2n+1}$$

is Convergent due to AST

At $x=1$, the PS is $\sum_{n=0}^{\infty} (-1)^n \frac{(1)^{2n+1}}{2n+1} = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}$

is Convergent due to AST.

Ex 10 ; Find the rational function of the PS $\sum_{n=1}^{\infty} nx^n$ for $|x|<1$ and evalueate $\sum_{n=0}^{\infty} \frac{n}{3^n}$

$$\underset{\text{rational } fu^n}{r(x)} = \left.\frac{P(x)}{q(x)}\right\} \text{ polynomial } q(x) \neq 0. \qquad \text{ex:- } \frac{x}{x^2+1}$$

$$\sum_{n=0}^{\infty} nx^n = \frac{\overset{\text{Polynomial}}{P(x)}}{\underset{\text{Polynomial.}}{q(x)}} \text{ with } q(x) \neq 0.$$

we know that ;

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad \text{for } |x|<1$$

Differentiating w.r.t x gives.

$$(1-x)^{-2} = \frac{d}{dx}\left[\sum_{n=0}^{\alpha} x^n\right] dx.$$

$$= \sum_{n=0}^{\alpha}\left[\frac{d}{dx}(x^n)\right] dx$$

$$(1-x)^{-2} = \sum_{n=0}^{\alpha} n x^{n-1}$$

$$\frac{1}{(1-x)^2} = \sum_{n=0}^{\alpha} n x^{n-1}$$

$$\Rightarrow \frac{x}{(1-x)^2} = \sum_{n=0}^{\alpha} n x^n = \sum_{n=0}^{\alpha} n x^n \quad \text{if } |x|<1$$

$$\sum_{n=1}^{\alpha} \frac{n}{3^n} = \sum_{n=0}^{\alpha} \frac{n}{3^n}$$

Replacing x by $1/3 < 1$, we gives

$$\frac{1/3}{(1-1/3)^2} = \sum_{n=1}^{\alpha} n\left(\frac{1}{3}\right)^n = \sum_{n=1}^{\alpha} \frac{n}{3^n}$$

$$\frac{1}{\cancel{3}_1} \times \frac{\cancel{9}\,3}{4}$$

$$\therefore \sum_{n=1}^{\alpha} \frac{n}{3^n} = 3/4$$

Theorem 04 :-

If $f(x) = \sum_{n=0}^{\infty} a_n (x-c)^n$ for $|x-c| < R$, then

$$f'(x) = \sum_{n=1}^{\infty} n\, a_n (x-c)^{n-1} \quad \text{for } |x-c| < R$$

$$f''(x) = \sum_{n=2}^{\infty} n(n-1)\, a_n (x-c)^{n-2} \quad \text{for } |x-c| < R.$$

$$f'''(x) = \sum_{n=3}^{\infty} n(n-1)(n-2)\, a_n (x-c)^{n-3} \quad \text{for } |x-c| < R$$

$\vdots$

$$f^{k}(x) = \sum_{n=k}^{\infty} n(n-1)(n-2)\ldots(n-(k-1))\, a_n (x-c)^{n-k} \quad \text{for } |x-c| < R$$

Also, we know that $n! = 1 \times 2 \times 3 \times \ldots \times (n-k)(n-(k-1)) \ldots (n-2)(n-1)n$

$$\Rightarrow n.(n-1)(n-2)\ldots(n-(k-1))$$

$$= \frac{n!}{1.2.3\ldots(n-k)}$$

$$= \frac{n!}{(n-k)!}$$

$$\therefore f^{(k)}(x) = \sum_{n=k}^{\infty} \frac{n!}{(n-k)!} a_n (x-c)^{n-k} \quad \text{for } |x-c| < R.$$